Venkata Lakshmi Parimala Pasupuleti

Founding Engineer · AI-Native Services · Production Agents · Evaluation Harness · HITL · Forward Deployed · Zero-to-One

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SUMMARY

Zero-to-one AI systems builder with 9+ years of production engineering depth. Built **Leasel** from a blank page: framed the problem, mapped the real workflow, built production agents with orchestration, tool use, durable state, deterministic compliance boundaries, and an evaluation harness - then shipped three customer-facing surfaces and operate them with a CI regression gate, post-launch monitoring, and audit logging on every agent decision. Built the human-review layer in: HITL escalation on every regulated decision, exception routing, full tracing - because keeping plausible-but-wrong AI output away from users is not optional. Built **AgentProbe** - open-source eval toolkit with golden datasets, pre-launch thresholds, regression suites, parallel validation runs, and post-launch monitoring across providers. At Magizhs (forward deployed), embedded with the client, mapped their workflow, and shipped a production agentic system against live data. At Entrata (9+ years, 100K+ users), owned a regulated compliance pipeline - the kind where a wrong output affects thousands of residents - and ran production incident response end to end, feeding every failure back into controls. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Production Agents · Agentic Workflows · Orchestration | Tool Use · Durable State · Deterministic Logic

Evaluation Harness · Regression Suites · Pre-Launch Thresholds | Human Review Layer · HITL · Exception Routing

Tracing · Audit Logging · Post-Launch Monitoring | Forward Deployed · Customer Workflow Mapping | Zero-to-One · Productization · Reusable Agents

LangGraph · LangChain · RAG Pipelines | Guardrails · Secure AI · Prompt Injection Defense | Responsible AI · Compliance Boundaries

Python · FastAPI · REST APIs | AWS · Docker · Kubernetes · CI/CD | SQL · PostgreSQL · System Integrations

Production Incident Response · Root-Cause Analysis | Regulated Domain · Compliance Workflows

EXPERIENCE

AI/ML Engineer (Volunteer)

Sep 2026 - Present

Blue Horse Digital — Virginia, USA

- Building ML models, AI-powered features, and data pipelines end-to-end; integrating AI components with backend APIs and surfacing insights into reporting layers for business stakeholders.

AI/ML Engineer Intern (Forward Deployed)

May 2025 - Aug 2025

Magizhs Tech Solution

- Embedded with client, mapped their real workflow, and built a production agentic system against live data - LangGraph stateful multi-agent pipeline with RAG, validation guardrails, hard compliance boundaries, and structured audit logging. Forward deployed from problem framing through production deployment.
- Deployed MCP server connecting agent outputs to downstream enterprise systems; defined integration contracts, handled authentication, built failure routing and exception paths; delivered FastAPI serving with CI/CD pipeline.

Senior Software Engineer

Sep 2018 - Aug 2024

Entrata - PropTech / Real Estate SaaS (100K+ monthly active users, regulated multi-tenant)

- **Owned a regulated compliance pipeline for nearly 7 years** - HUD income-limit ETL (federal rows → 180,000 domain records) powering eligibility and rent calculations for 100K+ housing units; zero compliance violations across ownership period; every release included validation, reconciliation, and rollback preparation because a wrong output affects thousands of residents.
- **Ran production incident response end to end** - owned root-cause analysis and postmortems; built internal debugging tools (40% debugging time reduction); drove corrective actions and documentation so failures could not recur; maintained bug count below 10 for 6+ consecutive months.
- **Built cross-functional stakeholder relationships** - collaborated directly with product, operations, compliance, and business teams; communicated technical tradeoffs clearly to non-technical stakeholders and leadership; delivered 20+ full-stack production modules across 120+ Agile sprints.
- PHI-compliant AWS S3 migration (100% coverage, zero security incidents); Python/Bash automation eliminating 80% of manual workload; microservices redesign (Docker) - 50% scalability improvement; mentored 5+ engineers; 300+ code reviews.

Associate Software Engineer

Jul 2017 - Sep 2018

Entrata India, Pune

- Built compliance and property management modules (Python, PHP, Java, PostgreSQL); led AWS S3 cloud migration; delivered large-scale data import and reconciliation workflows (1.6M records).

PROJECTS

Leasel - Zero-to-One Production Agentic Platform (Agents · Eval Harness · Human Review)

2026 | Python, LangGraph (8-agent), FastAPI, SQLite, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built from blank page: framed the problem, mapped the workflow, built production agents with orchestration, tool use, durable state, and deterministic compliance boundaries - three surfaces sharing one knowledge repository. **Human-review layer built in**: HITL escalation when agent confidence is low, exception routing on regulated decisions, full tracing on every agent action, audit logging on every decision. **Hard rule enforced in code**: LLM reasons and extracts; a human or deterministic check makes the compliance call - AI never auto-resolves regulated or financial outcomes.

- Evaluation harness: 1,240-case golden dataset, pre-launch thresholds, CI regression gate, LLM-as-judge scoring - 93.6% precision and 87.9% F1. Post-launch monitoring catches regressions before users see them.

AgentProbe - Evaluation Harness & Post-Launch Monitoring Toolkit 2025 | Python, FastAPI, LangGraph, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source eval toolkit built to operationalize the "keep plausible-but-wrong AI output away from users" requirement: golden datasets, pre-launch thresholds, regression suites, parallel validation runs against baseline, and post-launch monitoring. Tracing and structured logging across agent runs; 5-metric eval engine (accuracy, hallucination rate, retrieval quality, completeness, compliance); multi-provider routing (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama). Used as release control gate before every production deployment.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017